

Git, Part 2: The Humans

User Study Day · Do Developers Really Know How to Use Git Commands? A Large-Scale Study Using Stack Overflow

The Humans

Open the class link (it ends in /thu) and type your first name.

Do Developers Really Know How to Use Git Commands? A Large-Scale Study Using Stack Overflow

Yang, Zhang, Pan, Xu, Zhou, Huang · ACM TOSEM 2022

Take the paper's survey

The same questions it asked 92 developers.

Answer for yourself.

2. What is your professional area?
3. What is your highest degree?
4. How many years have you been using Git?
5. Which of the following levels do you consider your ability to use Git to be?
6. What are the main approaches you used to learn Git commands?
7. Do you have any suggestions for potential new developers on learning Git commands

Q1 of the form, the donation choice, is left out.

How the study was done

Collect	80,370 Stack Overflow questions about Git commands, 2008 to 2020.
Measure	Per command: how often viewed (popular), and how often no answer was accepted (hard).
Ask	A survey: 508 invited, 92 answered.

Who is in this study?

People watched while they used Git

People who asked on Stack Overflow, plus 92 surveyed

Git's own usage logs

1 · THE STUDY

People who asked. Nobody was watched.

People watched while they used Git

People who asked on Stack Overflow, plus 92 surveyed ✓

Git's own usage logs

Questions people asked, plus a survey.

Asking shows where people got stuck, not what they do.

What would watching show that asking can't?

When does a question count for `git reset`?

Someone reads it and decides

`git reset` appears in the question or its accepted answer

The asker tags it “reset”

1 · THE STUDY

A word search

Someone reads it and decides

git reset appears in the question or its accepted answer ✓

The asker tags it "reset"

Even if `git reset` appears only in the accepted answer, it counts.

Nobody reads the question to decide.

What could go wrong?

What they found

How many	80,370 questions: a steady 0.4% of Stack Overflow since 2010.
Who asks	Many long-time users: in 2020, 40% had joined Stack Overflow more than 5 years earlier.
Most viewed	Mostly undo: <code>revert</code> , <code>reflog</code> , <code>stash</code> , <code>clean</code> , <code>reset</code> .
Hardest	Rare commands; then <code>credential</code> and <code>submodule</code> .
How people learn	Mostly on their own: 81.7% of ticked answers.

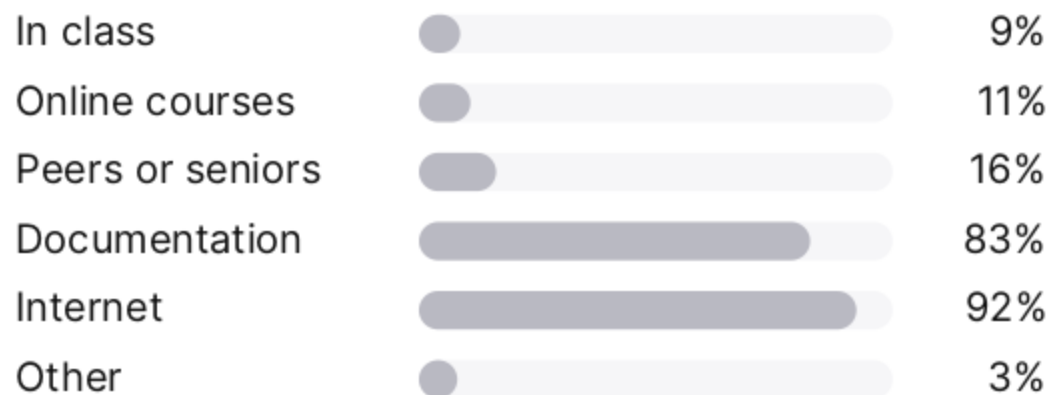
1 · THE STUDY

You and the paper's 92

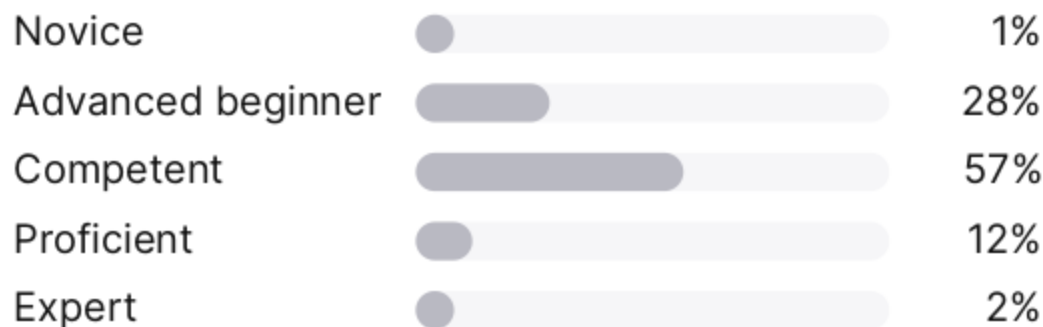
The 92 were found mostly through Stack Overflow.

Self-ratings and memories are shaky data.

THE PAPER'S 92: HOW THEY LEARNED (SHARE OF PEOPLE)



THE PAPER'S 92: GIT LEVEL, SELF-RATED



Median years using Git: 8. In class, the app shows your answers next to these.

Would you trust your own answers?

Read 8 real posts

Each was counted for one command.

Is the asker stuck on it?

STUCK ON IT

They ask about this command.

NEEDS IT, DOESN'T KNOW IT

It would fix their problem; they don't know it.

NOT ABOUT IT

It is only mentioned.

On your laptop: 8 posts, each with the question, the accepted answer, and the command the paper counted it for.

Drawn at random from short, answered posts credited to the top-5 commands (analysis/thursday_numbers.py, seed 294).

2 · CHECK IT YOURSELF

What the word search missed

In 5 of 8, the command is only in the answer.

The rule still counts all 8 as questions about it.

POST	COUNTED FOR	ASKER NAMES IT
git how to move back commit (6,163 views)	git reflog	Yes
How do I remove a single file from the staging area (undo git add)? (656,913 views)	git reset	Answer only
Undo a Git merge that hasn't been pushed yet (2,371,183 views)	git reflog	Answer only
How can i undo changes between two commits? (24 views)	git revert	Answer only
git 'merge' only existing files (1,378 views)	git clean	Yes
Git branch not working the way I expected it to (2,989 views)	git reset	Yes
revert a commit after push, checkout a commit behind and modification of files (46 views)	git revert	Answer only
Revert erroneous git checkout (45 views)	git stash	Answer only

Did they measure what's hard?

Sort 7 survey comments

The paper sorted comments into 6 groups.

Sort them yourself. Then compare.

The paper's six categories:

Learning and improving from practice (18)

Making the best use of the internet (17)

Understanding basic concepts and principles first (14)

Emphasizing the importance of Git (9)

Mastering only the basic commands (5)

Having the basic needs satisfied by Git GUI tools (2)

2 · CHECK IT YOURSELF

Did you agree?

The paper never says how it sorted, or if two people agreed.

Good practice: two people sort, then report agreement.

COMMENT	THE PAPER'S CATEGORY
#56 "Git is an extremely important tool to our routine as software ..."	Emphasizing the importance of Git
#32 "There are so many commands, I think it is better to learn them in ..."	Learning and improving from practice
#89 "Git is not a tool you have to go pretty deep to learn. Since it is a ..."	Mastering only the basic commands
#26 "The difficulty with git is that when problems arise, you have to dive ..."	Understanding basic concepts and principles first
#21 "All I want is to concentrate on my development, and dump the code ..."	Having the basic needs satisfied by Git GUI tools
#77 "Git is my VCS of choice, as I've learned how it works and have use ..."	Learning and improving from practice
#42 "Years on, I still constantly have to search the internet for help ..."	Making the best use of the internet

Would another team get the same groups?

BREAK

Break

5 minutes. Then groups.

Put one claim on trial

Your group gets one claim. One person types.

Write what they measured. Then what it really shows.

RQ2, §3.2

"This suggests that even developers with years of development experience can have trouble using Git commands."

§1, FINDING 4; §3.4

"...for the more frequently-used commands, git credential and git submodule are among the most difficult ones."

ABSTRACT; §3.5

"Self-learning is the primary learning approach."

§3.5

"This result, although surprising, is consistent with the conclusion we obtained in RQ2, indicating that even experienced developers still have doubts about Git usage."

3 · JUDGE IT

The verdicts (1/2)

The title asks if developers know how to use Git. Nobody was watched.

"This suggests that even developers with years of development experience can have trouble using Git commands."

THE DATA SUPPORTS

Some long-time users had trouble. True, as written.

"...for the more frequently-used commands, git credential and git submodule are among the most difficult ones."

THE DATA SUPPORTS

credential is above average (43.0% vs 36.5%);
submodule is about average (37.5%).

Which claim holds up best?

3 · JUDGE IT

The verdicts (2/2)

The title asks if developers know how to use Git. Nobody was watched.

"Self-learning is the primary learning approach."

THE DATA SUPPORTS

These 92 say they learned mostly from the internet and the docs.

"This result, although surprising, is consistent with the conclusion we obtained in RQ2, indicating that even experienced developers still have doubts about Git usage."

THE DATA SUPPORTS

Most of the 92 rate themselves competent or below (79 of 92).

Which claim holds up best?

Design a better undo

Undo is the most-viewed Git question.

Why do people have to ask? Design one fix, and a test.

YOUR FIX

What the user sees or types

HOW YOU WOULD TEST IT

Does it cut the time to recover for ...?

WHAT YOU MEASURE, AND

~~ONE RISK~~
e.g. time to recover; risk: people learn from the first try

HOW YOU COLLECT IT

Watch people, log real use, or ask them?

Your designs

Git added switch and restore in 2019. Did they help? Only a test with users can say.

Each group reads its change, its study, its measure and threat, and its data. The app shows all of them.

Which could you test by watching people?

END

What holds, what doesn't

Holds

- ✓ Git questions are many, and steady.
- ✓ Undo is a real need.
- ✓ Long-time users can struggle.
- ✓ The data is public. We checked it.

Doesn't hold, as written

- × "Hardest commands": fixes counted as problems.
- × The ranking: one post moves reflog from #8 to #2.
- × "Learn on their own": asked of people found on Stack Overflow.
- × The comment groups and "doubts": no method, self-ratings.
- × The title's question: nobody was watched.

END

2 min

One idea to take home

When you build a tool for people, ...

One line, on your own.

THURSDAY

Asking is not using.

Check what your data really records.